Wireless IoT Network Protocols

IoT Networking

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LoRaWAN®

The Internet of Things (IoT) is rapidly expanding, connecting billions of devices across diverse environments. Many of these applications require long-range communication with low power consumption, a niche perfectly filled by LoRaWAN. This introduction will delve into the intricacies of LoRaWAN technology, exploring its architecture, key features, and applications.

LoRa

- LoRa is a wireless modulation technique derived from Chirp Spread Spectrum (CSS) technology.
- It encodes information on radio waves using chirp pulses - similar to the way dolphins and bats communicate!
- LoRa modulated transmission is robust against disturbances and can be received across great distances.
- LoRa is ideal for applications that transmit small chunks of data with low bit rates. Data can be transmitted at a longer range compared to technologies like WiFi, Bluetooth or ZigBee
- These features make LoRa well suited for sensors and actuators that operate in low power mode.

LoRaWAN stands for Long Range Wide Area Network. It's a type of Low Power Wide Area Network (LPWAN) specifically designed for long-range communication with minimal energy usage.

LoRaWAN is a Media Access Control (MAC) layer protocol built on top of LoRa modulation. It is a software layer which defines how devices use the LoRa hardware, for example when they transmit, and the format of messages.

LoRaWAN is developed and maintained by the LoRa Alliance, ensuring interoperability between devices and networks from different vendors

LoRaWAN is suitable for transmitting small size payloads (like sensor data) over long distances. LoRa modulation provides a significantly greater communication range with low bandwidths than other competing wireless data transmission technologies.

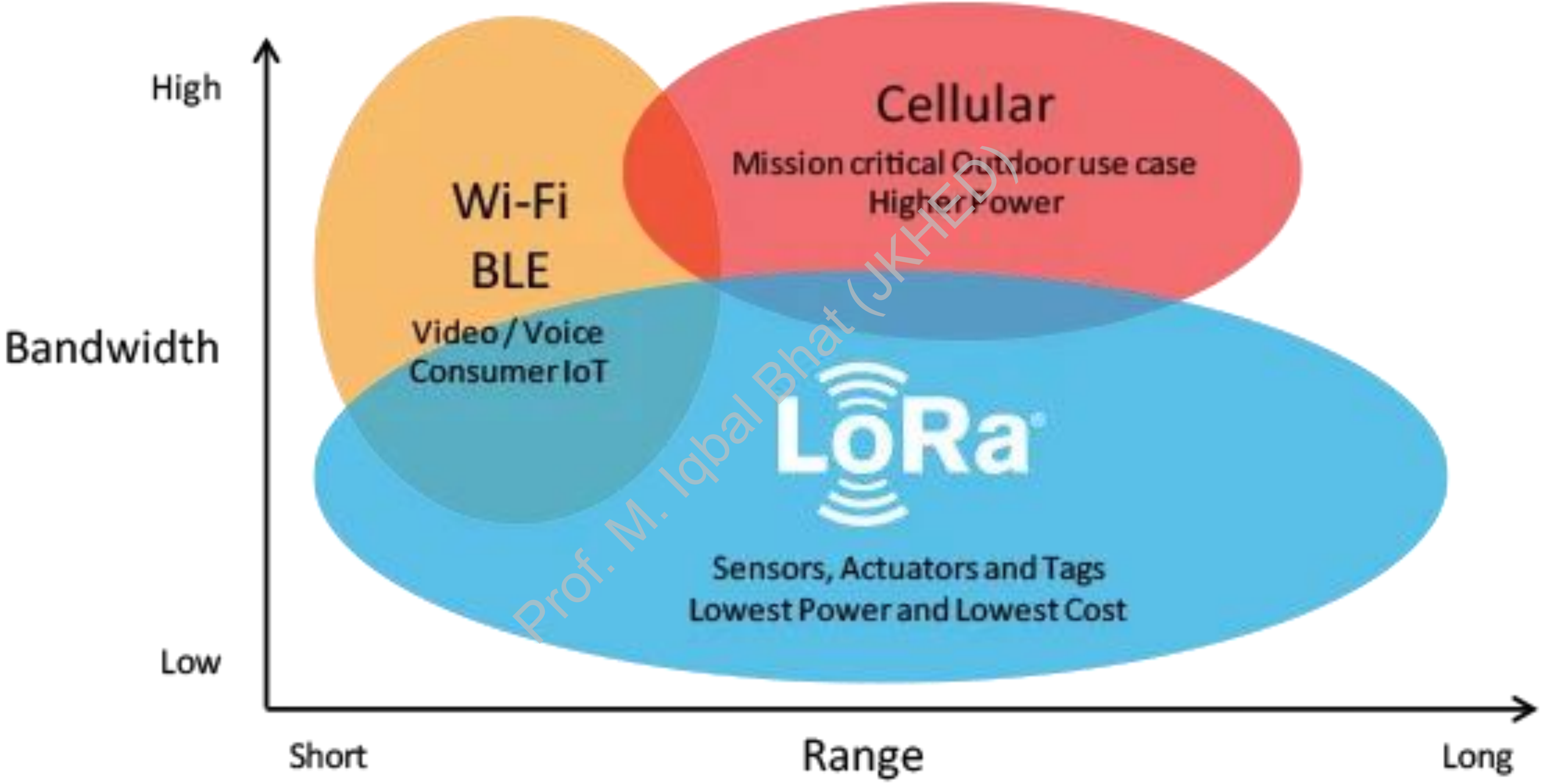
LoRa vs. LoRaWAN

LoRa

Refers to the physical layer modulation technique that enables long-range communication.

LoRaWAN

Defines the communication protocol and system architecture for LoRa-based networks.


High
Bandwidth
Low
Short
Range
Long
Wi-Fi
BLE
Video / Voice
Consumer IoT
Cellular
Mission critical Outdoor use case
Higher Power
LoRa
Sensors, Actuators and Tags
Lowest Power and Lowest Cost

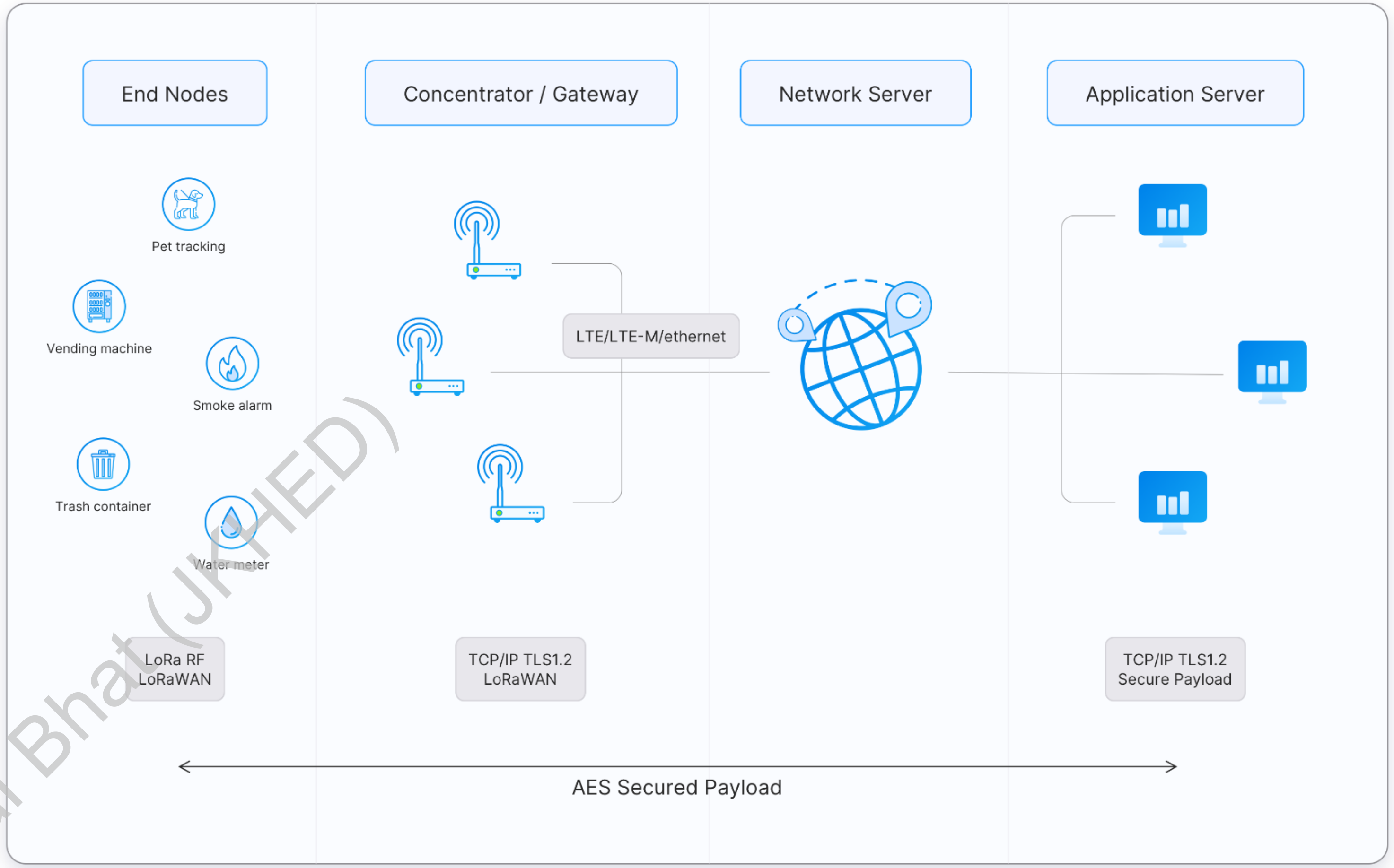
Features of LoRaWAN®

- **Ultra low power** - LoRaWAN end devices are optimized to operate in low power mode and can last up to 10 years on a single coin cell battery.
- **Long range** - LoRaWAN gateways can transmit and receive signals over a distance of over 10 kilometers in rural areas and up to 3 kilometers in dense urban areas.
- **Deep indoor penetration** - LoRaWAN networks can provide deep indoor coverage, and easily cover multi floor buildings.
- **License free spectrum** - You don't have to pay expensive frequency spectrum license fees to deploy a LoRaWAN network.
- **Geolocation**- A LoRaWAN network can determine the location of end devices using triangulation without the need for GPS. A LoRa end device can be located if at least three gateways pick up its signal.
- **High capacity** - LoRaWAN Network Servers handle millions of messages from thousands of gateways.
- **Low cost** - Minimal infrastructure, low-cost end nodes and open source software.
- **Certification program**- The LoRa Alliance certification program certifies end devices and provides end-users with confidence that the devices are reliable and compliant with the LoRaWAN specification.
- **Ecosystem**- LoRaWAN has a very large ecosystem of device makers, gateway makers, antenna makers, network service providers, and application developers.

LoRaWAN® Architecture

LoRaWAN® Architecture

- **End Devices** - sensors or actuators send LoRa modulated wireless messages to the gateways or receive messages wirelessly back from the gateways. .
- **Gateways** - receive messages from end devices and forward them to the Network Server.
- **Network Server** - a piece of software running on a server that manages the entire network.
- **Application servers** - a piece of software running on a server that is responsible for securely processing application data.
- **Join Server** - a piece of software running on a server that processes join-request messages sent by end devices



LoRaWAN® Security and Encryption

Secure Communication

LoRaWAN utilizes advanced encryption techniques to ensure secure data transmission between end-devices and the network. This includes encryption of the payload and authentication of devices to prevent unauthorized access.

End-to-End Encryption

LoRaWAN employs end-to-end encryption, where data is encrypted at the application layer before being transmitted over the network. This protects the confidentiality of the data, even if the network is compromised.

Device Authentication

Each LoRaWAN end-device is authenticated using unique keys, ensuring that only authorized devices can join the network. This prevents rogue devices from accessing the network and spoofing legitimate devices.

Secure Key Management

LoRaWAN has robust key management processes to securely distribute, update, and revoke encryption keys, minimizing the risk of key compromise and ensuring the continued security of the network.

LoRaWAN® Applications and Use Cases

Smart Agriculture

LoRaWAN enables low-power, long-range monitoring of soil, weather, and other agricultural data to optimize crop yields and resource usage.

Smart Cities

LoRaWAN supports a variety of IoT applications in urban environments, including traffic management, parking optimization, and environmental monitoring.

Healthcare

LoRaWAN-connected wearables and medical devices enable remote patient monitoring and continuous health data collection for improved care.

Industrial IoT

LoRaWAN helps optimize industrial operations by providing real-time insights into equipment status, energy usage, and environmental factors.

LoRaWAN® Advantages and Limitations

1

Long Range

Covers distances up to 10km in rural areas

2

Low Power

Devices can run for years on a single battery

3

Low Cost

Inexpensive hardware and network infrastructure

LoRaWAN offers several key advantages, including its long-range capabilities of up to 10km in rural settings, extremely low power consumption that enables multi-year battery life for devices, and a relatively low-cost hardware and network infrastructure. However, LoRaWAN also has limitations in terms of data throughput, latency, and security compared to other wireless technologies.

Conclusion

LoRaWAN presents a compelling solution for the growing demand for long-range, low-power connectivity in the IoT landscape. Its open standard, scalability, and security features make it a versatile technology for a wide range of applications across various industries. Understanding LoRaWAN's capabilities and limitations is crucial for developers and businesses looking to leverage its potential for their specific needs.

The Internet of Things (IoT) encompasses a vast array of devices with varying capabilities and constraints. While IPv6 offers a robust and scalable addressing scheme for the internet, its large packet sizes and header overhead pose challenges for resource-constrained devices common in low-power wireless networks. This lecture will explore 6LoWPAN, a technology designed to adapt IPv6 for these networks, enabling seamless integration of low-power devices into the Internet of Things.

Introduction to 6LoWPAN

6LoWPAN stands for IPv6 over Low power Wireless Personal Area Networks.

It defines a set of mechanisms and protocols to enable efficient transmission of IPv6 packets over resource-constrained networks, such as those based on IEEE 802.15.4.

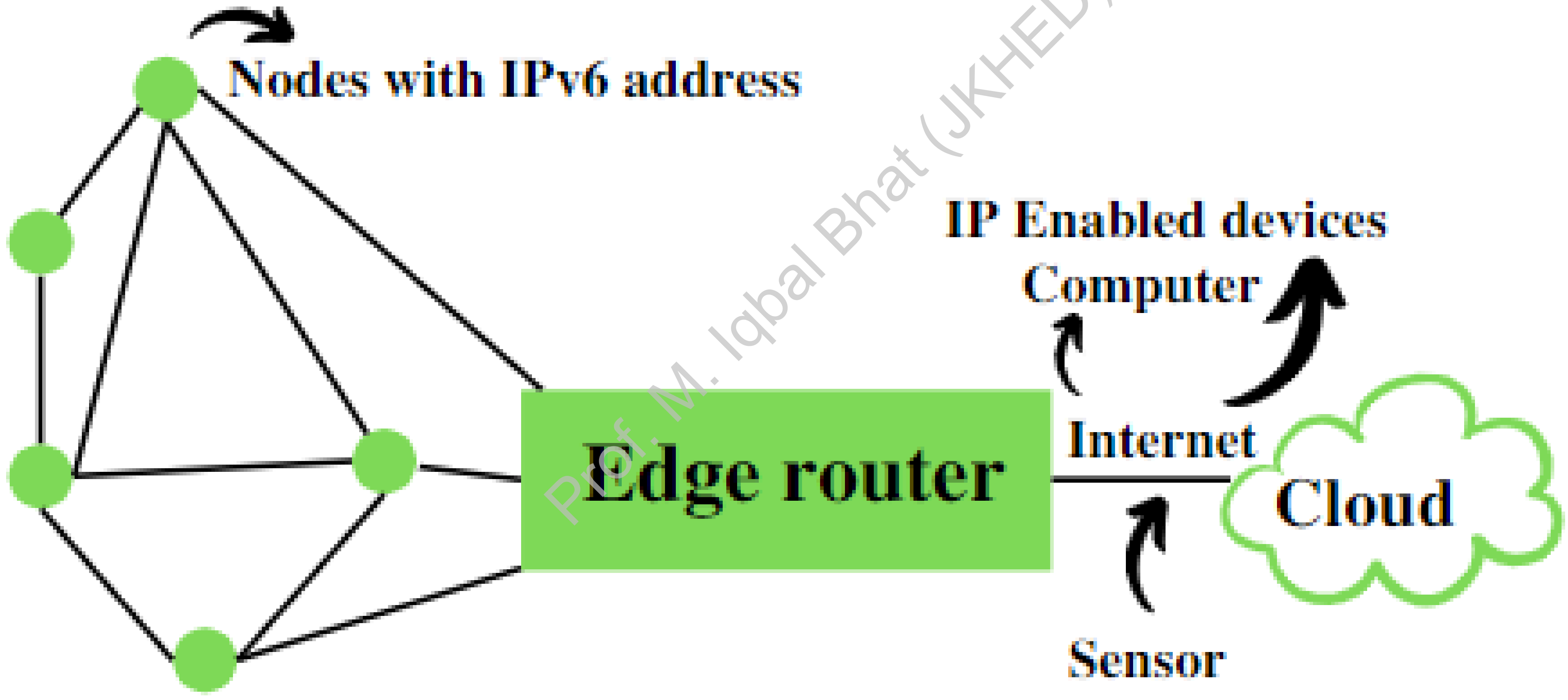
Adaptation Layer: 6LoWPAN operates as an adaptation layer between the network layer (IPv6) and the data link layer (IEEE 802.15.4), performing header compression, fragmentation, and other optimizations to adapt IPv6 packets for transmission over low-power wireless links.

It uses AES 128 link layer security, which AES is a block cipher having key size of 128/192/256 bits and encrypts data in blocks of 128 bits each. This is defined in IEEE 802.15.4 and provides link authentication and encryption.

Features of 6LoWPANs

- ✓ Allows IEEE 802.15.4 radios to carry 128-bit addresses of Internet Protocol version 6 (IPv6).
- ✓ Header compression and address translation techniques allow the IEEE 802.15.4 radios to access the Internet.
- ✓ IPv6 packets compressed and reformatted to fit the IEEE 802.15.4 packet format.
- ✓ Uses include IoT, Smart grid, and M2M applications.

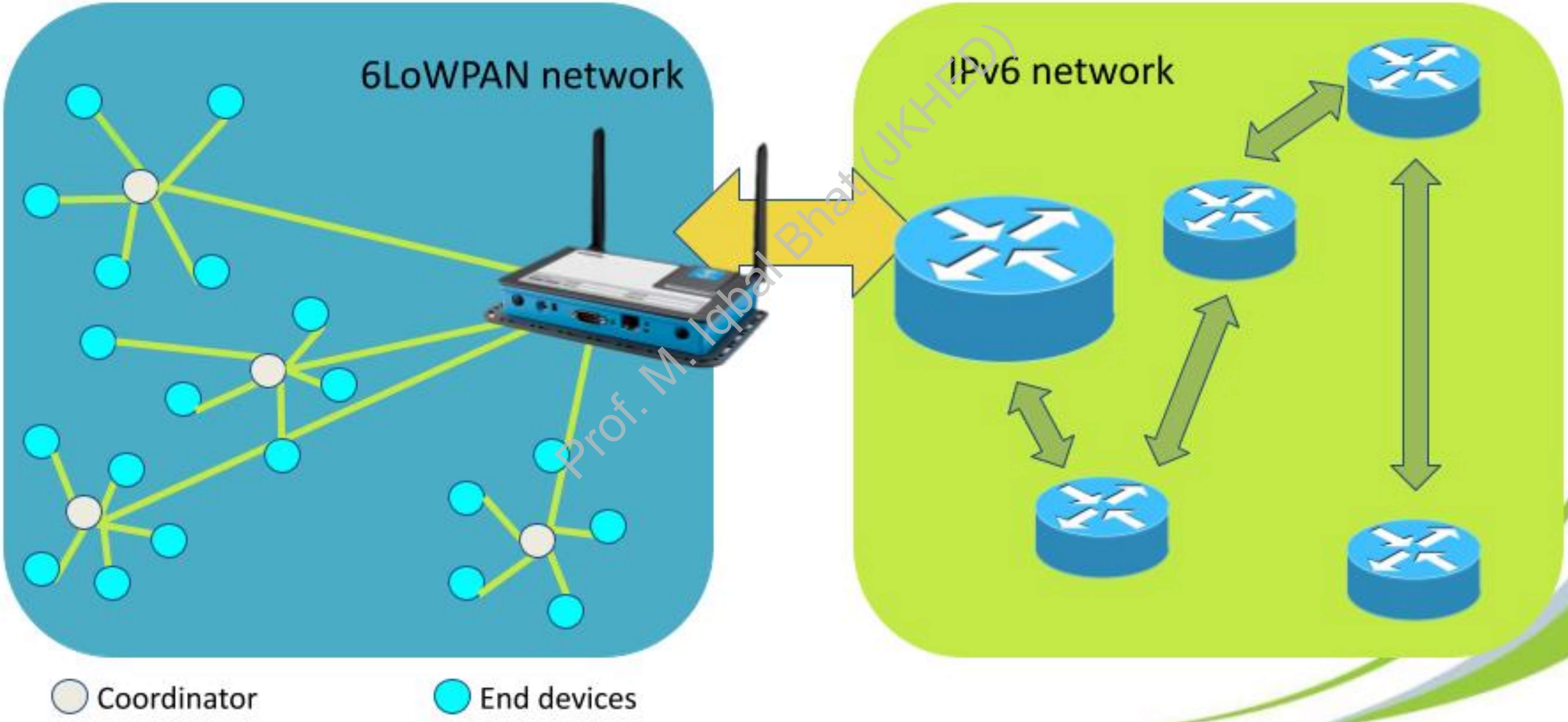
6LoWPAN Architecture



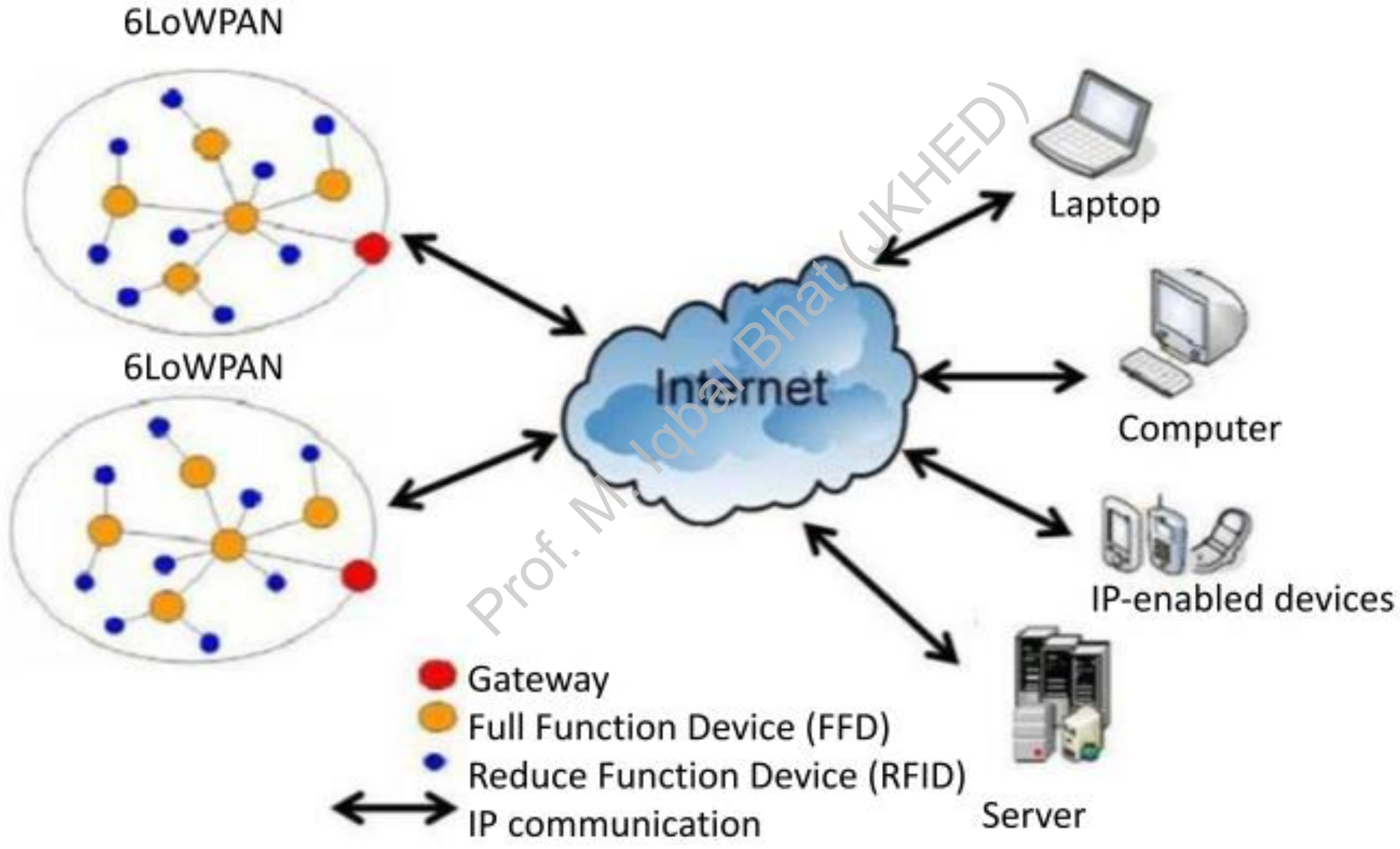
6LoWPAN Architecture

1. **6LoWPAN Devices:** These are the heart of the network, encompassing various resource-constrained devices with limited power budgets. Examples include sensors, actuators, wearables, and smart home appliances. These devices can be further categorized as:
    1. **Hosts (End Devices):** These devices primarily collect or transmit data and lack the capability to route it for other devices. They operate in a low-power mode, periodically waking up to exchange data with the router.
    2. **Routers:** As the name suggests, these devices route data packets within the 6LoWPAN network, ensuring it reaches the intended recipient. Routers typically have more processing power and battery capacity compared to hosts.
2. **Edge Router:** This is the bridge between the 6LoWPAN network and the wider internet (or other IPv6 networks). It performs three crucial functions:
    1. **Data Exchange:** It facilitates the exchange of data between devices within the 6LoWPAN network and the internet or other external networks.
    2. **Local Data Routing:** It handles the routing of data packets between devices solely within the 6LoWPAN network.
    3. **Radio Subnet Management:** It oversees the creation and maintenance of the 6LoWPAN network, which acts as a radio subnet within the broader network infrastructure.
3. **Access Point (AP):** While not directly part of the 6LoWPAN network itself, the AP serves as the connection point for devices within the broader network, typically including PCs, servers, and other internet-connected devices. The AP often functions as an IPv6 router, facilitating communication between these devices and the internet.

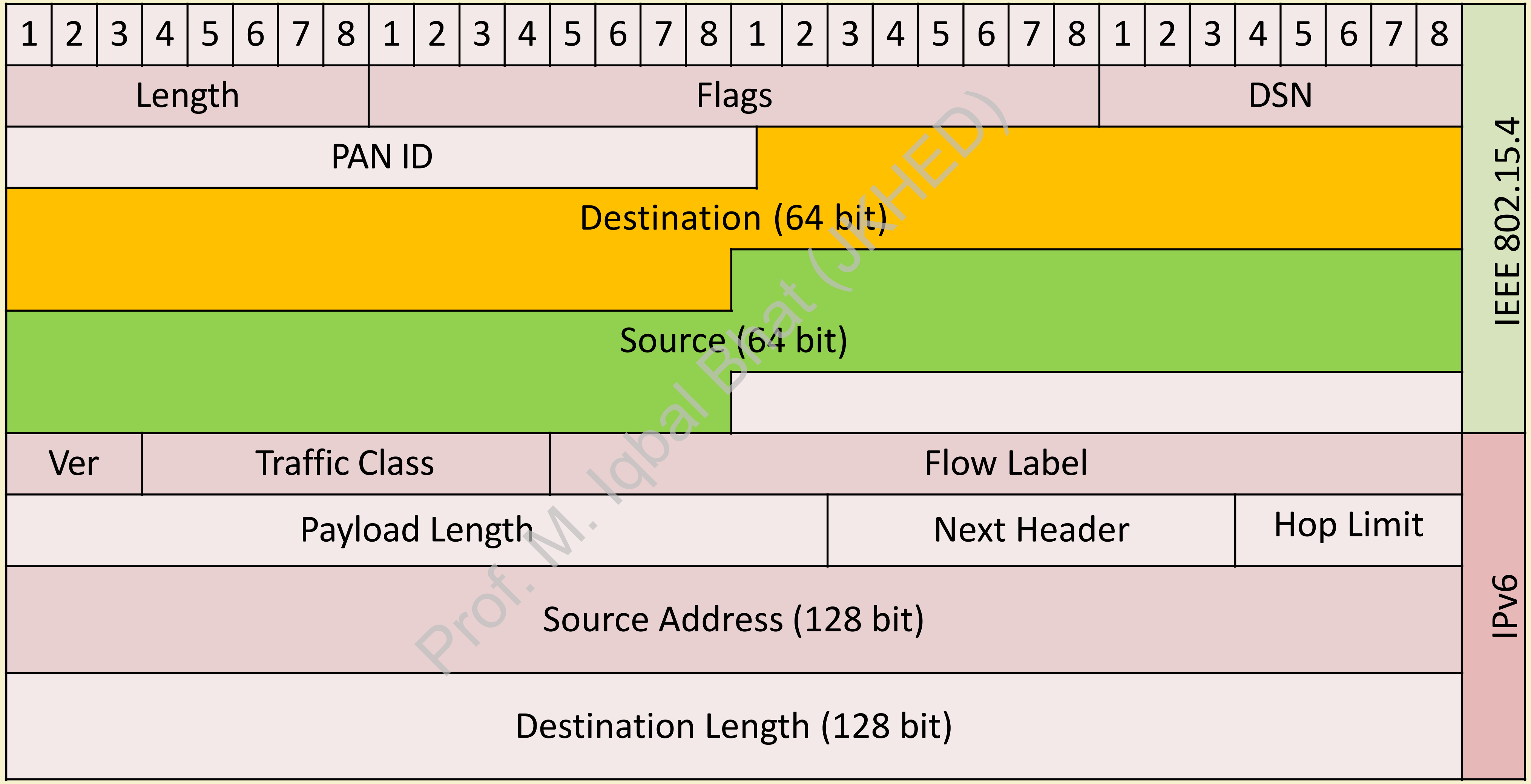
6LoWPAN network model



6LoWPAN Architecture

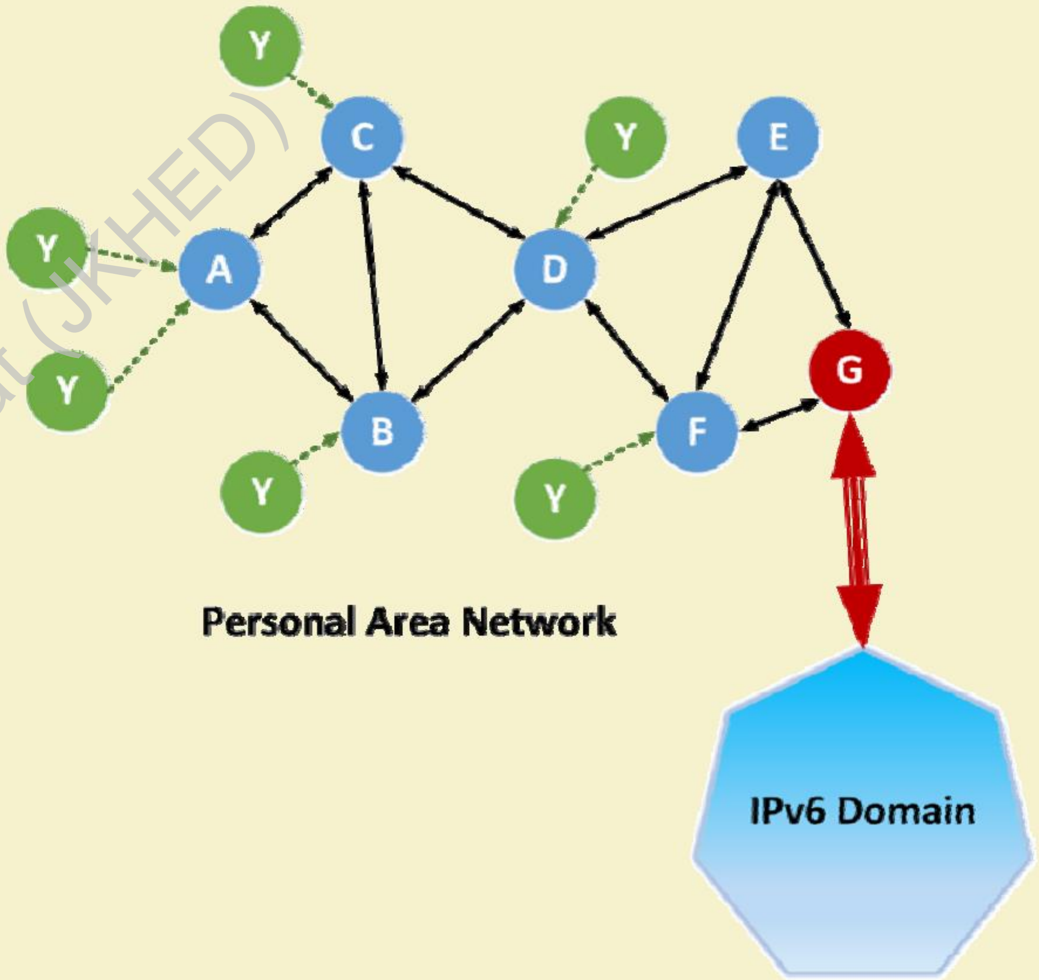


6LowPAN Packet Format



6LoWPAN Routing Considerations

- ✓ Mesh routing within the PAN space.
- ✓ Routing between IPv6 and the PAN domain
- ✓ Routing protocols in use:
  - **LOADng**
  - **RPL**



Benefits of 6LoWPAN

1. **IP Connectivity**: 6LoWPAN enables direct IP connectivity for low-power devices, allowing them to communicate with any other device on the internet without the need for gateways or protocol translations.
2. **Interoperability**: By utilizing IPv6, 6LoWPAN ensures interoperability between devices from different vendors and promotes seamless integration into existing IP infrastructure.
3. **Scalability**: The use of IPv6 provides ample address space to accommodate the massive number of devices expected in the IoT.
4. **End-to-End Security**: 6LoWPAN benefits from the security features of IPv6, enabling secure communication and data protection for low-power devices.

Applications of 6LoWPAN

1. **Smart Homes and Buildings**: Building automation, smart lighting, HVAC control, and energy management systems.
2. **Industrial Automation**: Wireless sensor networks for monitoring and control in industrial environments.
3. **Smart Agriculture**: Precision agriculture applications like environmental monitoring and irrigation control.
4. **Environmental Monitoring**: Monitoring air quality, water quality, and other environmental parameters.
5. **Asset Tracking**: Tracking the location and status of assets in various industries.

Conclusion

6LoWPAN plays a crucial role in bridging the gap between low-power wireless networks and the IP world, enabling seamless integration of resource-constrained devices into the Internet of Things. Its efficient adaptation of IPv6 and support for mesh networking make it a valuable technology for a wide range of IoT applications, promoting interoperability, scalability, and end-to-end connectivity. As the IoT continues to evolve, 6LoWPAN is poised to play an increasingly significant role in connecting the billions of devices that will shape our future.